CS1200 — Discrete Mathematics for Computer Science

Tier	A
Phase (order)	1
Do this first	School maths only
Estimated effort	10-12 weeks, ~6 h/week
Unlocks	CS1202, CS2200, CS2800, CS6030

Roadmap

1. Logic first: propositional and predicate logic, truth tables, resolution.
2. Then proofs: direct, contradiction, induction (weak and strong). Spend the most time here; every later theory course is proofs.
3. Sets, relations, equivalence relations, functions, countable vs uncountable (Cantor).
4. Counting: permutations, combinations, inclusion-exclusion, pigeonhole, recurrences.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Computer Sc - Discrete Mathematical Structures (40 videos). NPTEL *Discrete Mathematical Structures* (Kamala Krithivasan, IIT Madras): closest to IITM's own syllabus.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Discrete Mathematics (100+ videos). NPTEL-NOC IITM *Discrete Mathematics* (Sudarshan Iyengar, IIT Ropar): very beginner-friendly, puzzle-style.
- MIT 6.042J Mathematics for Computer Science, Fall 2010 (25 videos). MIT 6.042J (Fall 2010, Tom Leighton): proofs, induction, counting, probability. Pairs with the free MCS PDF.
- MIT 6.1200J Mathematics for Computer Science, Spring 2024 (24 videos). MIT 6.1200J (Spring 2024): the newest version of 6.042.

⚡ Short videos for intuition and revision

- Discrete Mathematics (100+ videos). Neso Academy *Discrete Mathematics*: short revision videos.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1-2	1–9	Propositional + predicate logic, inference, resolution, proof methods. Do 20 Rosen ch.1 exercises.
3	10–12	Sets and induction. Write 10 induction proofs (sums, divisibility, inequalities) on paper.
4-5	13–23	Relations, graphs/trees, closures, partial orders, equivalence relations and partitions.
6	24–27	Functions, countability, pigeonhole principle.
7-8	28–34	Permutations/combinations, generating functions, recurrence relations. Solve recurrences by hand and check in Python.
9	35–40	Algebras, finite automata, lattices (preview of CS1202/CS2200).

▼ **All lectures in *Computer Sc - Discrete Mathematical Structures* (40 videos)**

1. Lecture 1 - Propositional Logic
2. Lecture 2 - Propositional Logic (Contd.)
3. Lecture 3 - Predicates & Quantifiers
4. Lecture 4 - Predicates & Quantifiers (Contd.)
5. Lecture 5 -Logical Inference
6. Lecture 6 -Resolution Principles & Application to PROLOG
7. Lecture 7 - Methods of Proof
8. Lecture 8 - Normal Forms
9. Lecture 9 - Proving programs correct
10. Lecture 10 - Sets
11. Lecture 11 - Induction
12. Lecture 12 - Set Operations on Strings Over an Alphabet
13. Lecture 13 - Relations
14. Lecture 14- Graphs
15. Lecture 15 - Graphs (Contd.)
16. Lecture 16 - Trees
17. Lecture 17 - Trees and Graphs
18. Lecture 18 - Special Properties of Relations
19. Lecture -19 - Closure of Relations
20. Lecture 20 - Closure of Relations (Contd.)
21. Lecture 21 - Order Relations
22. Lecture 22 - Order and Relations and Equivalence Relations
23. Lecture 23 - Equivalence relations and partitions
24. Lecture 24 - Functions
25. Lecture 25 - Functions (Contd.)
26. Lectures 26 - Functions (Contd.)

27. Lecture 27-Pigeonhole Principle
28. Lecture 28 - Permutations and combinations
29. Lecture 29 - Permutations and Combinations(Contd...)
30. Lecture 30 - Generating Functions
31. Lecture 31- Generating Functions
32. Lecture 32 - Recurrence Relations
33. Lecture 33 - Recurrence Relations
34. Lecture 34-Recurrence Relations(contd..)
35. Lecture 35-Algebras
36. Lecture 36-Algebras(contd...)
37. Lecture 37-Algebras(contd...)
38. Lecture 38-Finite State Automaton
39. Lecture 39-Finite State Automaton(contd....)
40. Lecture 40-Lattices

Books, notes and websites

- ✅ NPTEL *Discrete Mathematical Structures* (Kamala Krithivasan, IIT Madras) — nptel.ac.in/courses/106106094
- ✅ NPTEL *Discrete Mathematics* (Sudarshan Iyengar, IIT Ropar) — nptel.ac.in/courses/106106183
- MIT 6.042J *Mathematics for Computer Science* (Lehman, Leighton, Meyer) — free PDF + OCW videos
- Book: Rosen, *Discrete Mathematics and Its Applications* (exercise bank)

Practice / proof of understanding

Prove 10 statements by induction and 10 by contradiction, writing each proof in full sentences. Solve the MIT 6.042J problem sets.

Note

You can run this in parallel with CS1111; it needs no programming.

What your IITM course sheet says

From the PDF: CS1200

Official description: The course is about sets which are discrete in nature. It starts from ways of describing them, then size estimates of sets (combinatorics), and then culminates in studying structured sets. After completing the course, the student will be able to:(1) Use logical notation to define and reason mathematically about the fundamental data types and structures (such as numbers, sets) used in computer algorithms and systems. (2) Comprehend and Evaluate rigour in the definitions and conclusions about mathematical models and identify fallacious reasoning and statements. (3) Identify and Apply properties of combinatorial structures and properties - know the basic techniques in combinatorics and counting.(4) Analyze sets with operations, and identify their structure. (5) Reason and Conclude properties about the structure based on the observations. (6) Gain the conceptual background needed to

Official syllabus (trimmed):

> quences: Data Models. - 3L+1T (1W) Finite Sets, Power Set, Cardinality of finite sets, Cartesian Product, Properties of Sets, Vector Implementations of Sets. Describing Sets : Logic & Proofs - 15L + 5T (5W) Introduction to Logic. Propositional Logic, Truth tables, Deduction, Resolution, Predicates and Quantifiers, Mathematical Proofs. Infinite sets, well-ordering. Countable and Uncountable sets, Cantor's diagonalisation. Mathematical Induction - weak and strong induction. Relational Structures on Sets : Relations & Graphs - 9L + 3T (3W) Relations, Equivalence Relations. Functions, Bijections. Binary relations and Graphs. Trees (Basics). Posets and Lattices, Hasse Diagrams. Fixed points and their computation. Knaster-Tarski Fixed point theorem. Boolean Algebra. Sizes of Sets : Counting & Combinatorics - 15L + 5T (5W) Counting, Sum and product rule, Principle of Inclusion Exclusion. Pigeon Hole Principle, Counting by Bijections. Double Counting. Linear Recurrence relations - methods of solutions. Generating Functions. Permutations and counting.

Official textbooks: Discrete Mathematics and its Applications – Kenneth H. Rosen 7th Edition -Tata McGraw Hill Publishers - 2007.

Official references: 1) Elements of Discrete Mathematics, C. L Liu, McGraw-Hill Inc, 1985 2) Concrete Mathematics, Ronald Graham, Donald Knuth, and Oren Patashnik, 2nd Edition - Pearson Education Publishers - 1996 3) Combinatorics: Topics, Techniques, Algorithms by Peter J. Cameron, Cambridge University Press, 1994 (reprinted 1996). 4) Applied Combinatorics, Alan Tucker, 6th edition, Wiley, 2012.